Assignment 1

Name: _____ Student ID: _____ Marks: 40

Apart from the geometric applications such as computing volumes and finding areas of surfaces, double integrals have several physical applications such as computing mass, electric charge, center of mass, and moment of inertia. Discuss the concept of double integrals on these four physical applications. For each application, find at least two examples and solve them using double integrals. [40]

Note : Please note that the presentation of solutions (clarity, coherence, conciseness, and completeness) is important. Show your work and organize your solution. An answer without proper justification may receive less or even zero credit.